

Aryamaan (AJ) Jena

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Education

UCLA Anderson School of Management

PhD in Management (Finance); Fink Center Award, Anderson Departmental Fellowship Los Angeles, CA 2025 - 2030 (expected)

• Coursework: Asset Pricing Theory, Empirical Asset Pricing, Empirical Corporate Finance, Econometrics, Machine Learning

University of Chicago

MS in Financial Mathematics (Financial Data Science); ASEAN Merit Scholarship Chicago, IL 2021 - 2022

SP Jain School of Global Management

BS in Data Science; Merit Scholarship; Dean's List (4 times) Mumbai and Sydney 2018 - 2021

Research

Conditioning or Orientation? Shrinkage Targets for Characteristic-Managed Portfolios Working paper, 2026

- Proposed the second moment of firm characteristics as a covariance shrinkage target for 132 characteristic-managed portfolios (JKP data, 1963-2025).
- Highest out-of-sample Sharpe ratio of 23 estimators, but statistically tied with a scaled-identity target; a rotation test shows most of the gain comes from regularization, not characteristic information.

Returns on Presale Housing Markets in India

Working paper, 2025

- Repeat-sales estimates of returns on presale housing in Bangalore, NCR, Chennai and Pune.

Research Experience

Indian Institute of Management Bangalore

Research Associate, Prof. Venkatesh Panchapagesan Bengaluru, India 2024 - 2025

- Built housing price datasets for four Indian cities and estimated presale returns (working paper above).

Harvard Business School

Research Associate, Profs. Victoria Ivashina and Josh Lerner Boston, MA 2023 - 2024

- Ran mean-variance and Black-Litterman simulations in Python to estimate the cost of illiquidity in private equity and real estate allocations.
- Built a dataset of US financial firms and ran synthetic control regressions on how acquisitions affect FinTech innovation.

University of Chicago Booth School of Business

Research Assistant, Prof. Dacheng Xiu Chicago, IL 2022 - 2023

- Predicted Zillow pricing errors with linear, regularized and nonparametric ML regressions.
- Built LendingClub default prediction models, later turned into assignments and lecture material for MBA students.

Industry Experience

Cboe Global Markets, Cboe Labs

Derivatives Strategy Analyst Intern Chicago, IL Jun - Dec 2022

- Wrote a filtering algorithm for the VIX calculation that uses put-call parity bounds to flag anomalous option quotes causing spikes in the index; built it as a tested, object-oriented Python module in the firm-wide quant library.
- Built a Python library for SEC EDGAR filings, automated iBoxx index compliance tests, moved workflows to AWS.
- Received a full-time return offer.

Pareto Frontier Capital

Quantitative Research Intern Chicago, IL Oct 2021 - Jan 2022

- Feature engineering and PCA on market and on-chain data for crypto trading signals.

Teaching

UCLA Anderson, TA: Credit Markets (Prof. Julien Pénasse); Venture Capital and Private Equity

Fall 2026

Chicago Booth, TA: Financial Econometrics (Prof. Jeffrey Russell); Decoding FinTech (Prof. Dacheng Xiu)

2022

Projects

Prediction market price discovery

2026

- Matched 759 contracts listed on both Polymarket and Kalshi using text matching, then measured price gaps, how fast they close, and which venue leads price discovery.

Skills

Methods: Covariance shrinkage, factor models, regularized regression, gradient boosting, PCA, synthetic control

Tools: Python (pandas, scikit-learn, statsmodels, LightGBM), R, SQL, Git, AWS, LaTeX; WRDS (CRSP, Compustat, JKP)